Transition design: Co-creating system solutions for chronic obstructive pulmonary disease (COPD) care

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Choosing the appropriate design process is critical for the effective implementation and long-term sustainment of interventions aimed at addressing public health challenges. To address this need, we proposed a Transition Design model to identify and deliver sustainable solutions for complex healthcare problems. This model generates system-level health-intervention initiatives that can synergistically function, particularly during the development and implementation phases, to enhance healthcare delivery. Drawing from a case study on addressing the challenges of hospitalisations and early discharge for people with acute exacerbations of chronic obstructive pulmonary disease (AECOPD), we reflected on the process and analysed the outcomes. Although further testing for the initiatives is warranted, this study contributes to the evolving discourse in design research on systems solutioning for public-health challenges.
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1 Introduction

The inherent complexity of public health problems is increasingly being recognised, and this is reflected in the design of interventions that are intended to address them (Craig et al., 2008). Implementing and sustaining these interventions often presents challenges (Aarons, Hurlburt, & Horwitz, 2011; Craig et al., 2008). Sustainment, which is the final stage of the implementation process, involves creating and supporting the structures and processes that allow the continued use of implemented programmes by systems or organisations (Aarons et al., 2011, 2016; Chambers, Glasgow, & Stange, 2013). However, the constrained timeframes within research, which are set by funding systems, as well as the financial and political pressure on decision-makers to adopt short-term perspectives, limit research on how to design public health interventions for both effective implementation and long-term sustainment (Aarons et al., 2014; Chambers et al., 2013).

As the challenges facing public healthcare grow in complexity, the constraints of conventional design approaches become apparent (Jones, 2013); these often focus on immediate solutions within constrained settings, and potentially overlook diverse stakeholder needs (Irwin, 2020). This complexity contributes to significant implementation barriers, with estimates suggesting that over 85 % of expenditure on medical research has a minimal impact on public health due to ineffective implementation (Chalmers & Glasziou, 2009).

In response to these challenges, we developed a Transition Design model, recognising that various iterations of models based on the Transition Design methodology have been created and used according to specific contexts. Our model integrates co-creation principles and strategies for fostering appreciative intelligence in public health contexts. We adopted co-creation principles because they facilitate complex interactions between diverse stakeholders, ensuring that all perspectives are actively engaged in the process (Leask et al., 2019). In addition, we integrated appreciative intelligence into the Transition Design methodology, as it has been shown to strengthen stakeholder involvement and promote active, meaningful inputs (Ghaye et al., 2008).

This paper contributes to the evolving discourse in design research on systems solutions for public health challenges. We believe that our integrated model provides a deeper understanding of the complexities inherent in public health problems, which could enhance the implementation of interventions and strengthen healthcare systems over time.

In this paper, we 1) delineate the rationale and methodology behind the integration of a Transition Design model in our research, 2) introduce our model, which is the design methodology used in this study, 3) present the complexities associated with hospitalisation issues in chronic obstructive pulmonary disease (COPD) care, 4) share insights from our case study on COPD hospitalisation, and 5) discuss the implications of this research for other design researchers who are investigating applying systems thinking to health contexts.

2 *Rationale for introducing a Transition Design model when tackling public health problems*

The difficulties involved in implementing public health interventions are closely tied to the ways in which these interventions are designed. One major concern regarding the process of designing such interventions is the insufficient alignment of stakeholder interests, including patients and healthcare practitioners, which can lead to unsuccessful implementation (An, Sandlund, et al., 2023; Nair, Andersson, Nygren, & Lundgren, 2023). The complex interactions between users, other stakeholders, and technology during implementation highlight the need for a more integrated design approach (An, Sandlund, et al., 2023; Greenhalgh & Papoutsi, 2018). Additionally, there is often a lack of empirically grounded theoretical foundation and guiding principles for designing interventions that promote long-term sustainment (Aarons et al., 2014; Chambers et al., 2013).

Acknowledging that public health issues are frequently embedded within complex, interconnected problem clusters in the broader healthcare system is crucial for enhancing the design process (Kiekens, Dierckx de Casterlé, & Vandamme, 2022). In this context, the term 'public health issues' refers specifically to persistent, long-term challenges within health systems that are difficult to address. These entrenched problems demand comprehensive, sustained efforts in order to be effectively managed and resolved. Addressing these issues requires recognising that leverage points may not always lie where we assume they are, as systems often operate counterintuitively (Forrester, 1971; Meadows & Wright, 2008). Here, leverage points refer to specific areas within a health system where small changes in practices can lead to significant, system-wide improvements (Abson et al., 2017). Systems thinking offers a methodology for uncovering these leverage points, providing structural explanations for the root causes of problems, and suggesting long-term strategies for enhancing system performance (Payne, 2022). While systems thinking reveals the linkages, interdependencies, and complexity of our surroundings, it is through the discipline of design that action can be taken (Buchanan, 2020). In this study, Transition Design, as a systems-thinking methodology, is suggested as a foundation for identifying the leverage points within health systems.

Transition Design represents an evolution of both service design and design for social innovation, building on foundational principles while pushing the boundaries of what design can achieve (Irwin, 2015). Service design, which emerged in the 1980s, adopts a systems approach and participatory method, and laid the groundwork for Transition Design (Irwin, 2015). Similarly, design for social innovation embraces systems thinking, and aims to create solutions that challenge existing socioeconomic and political paradigms (Irwin, 2015). What distinguishes Transition Design from its predecessors is that it facilitates design being undertaken within radically new socio-economic and political paradigms (Irwin, 2015). Its adaptability has made Transition Design an effective tool in addressing a wide range of complex, interrelated problems. For example, Transition Design has been utilised to address Latin America's multiple complex social, economic, and environmental challenges (Juri, Zurbriggen, Bosch Gómez, & Ortega Pallanez, 2021), the water-shortage problem in Ojai (Irwin, 2020), and difficulties in Japan relating to lifestyle and careers (Iwabuchi & Mizuno, 2020). By addressing such diverse issues, the potential of Transition Design to drive systemic change across a variety of contexts has been demonstrated, as has its ability to offer solutions that align with new, sustainable paradigms.

We believe that Transition Design offers a powerful framework for tackling the complexity of public health problems due to several key factors. By adopting a systems-thinking approach, Transition Design enables a comprehensive understanding of the intricate nature of these challenges, avoiding fragmented solutions that often yield limited results in healthcare systems (Jarrell et al., 2024). Its support for multidisciplinary collaboration further enhances its ability to address the multifaceted nature of modern health challenges, fostering comprehensive and sustainable solutions through cross-sector engagement (Feo, Urry, Conroy, & Kitson, 2023; Irwin, 2020). Furthermore, Transition Design's adaptability allows it to evolve in response to shifting circumstances, offering the flexibility needed to address emerging problems effectively (Irwin, 2020). Moreover, it not only emphasises the development of long-term, sustainable solutions, but prioritises stakeholder involvement, empowering them in the co-creation of interventions (Irwin, 2020). This approach aligns with the current healthcare landscape, where the active participation of diverse players is essential.

Transition Design provides a structured framework consisting of three core phases: reframing the present and future (i.e. mapping the current problem and envisioning a long-term future in which the issue is resolved), designing interventions (i.e. linking and amplifying projects aimed at problem resolution and systems transition), and waiting and observing (i.e. understanding how a system responds to interventions) (Figure 1). These phases guide stakeholders in co-creating visions of desirable futures and identifying leverage points within larger systems, wherein an "ecology of actions" can be implemented (Irwin, 2020). As the output of Transition Design when addressing complex

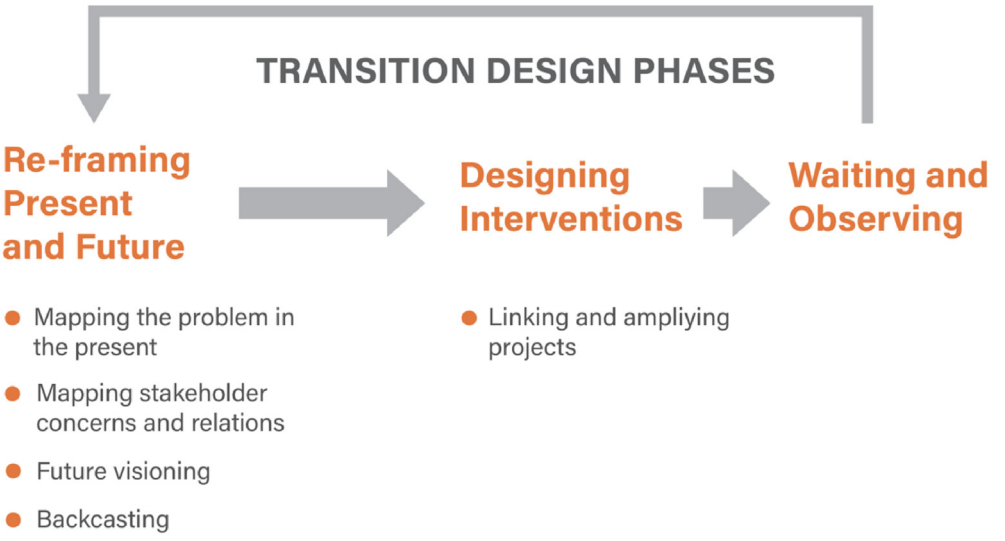


Figure 1 The phases of Transition Design (Irwin, 2020)

public health problems, an "ecology of actions" (Irwin, 2015) here in this study refers to system-level health interventions. We define system-level health interventions as a collection of interventions that complement one another in both the development and implementation phases, and that can be used together to improve the provision of healthcare (Irwin, 2015), based on the idea that 'the whole is greater than the sum of its parts'.

Co-creation can be a useful complement to systems-thinking methodologies such as Transition Design (Smith, Georgiou, Jalali, & Chastin, 2024). Its primary strength is suggested to be that it may enhance the implementation processes of public health interventions by actively involving stakeholders and facilitating meaningful interactions (Leask et al., 2019; Pérez Jolles et al., 2022; Voorberg, Bekkers, Timeus, Tõnurist, & Tummers, 2017). Emerging as a key approach in health research, co-creation is defined as a collaborative approach to creative problem-solving between diverse stakeholders at different stages of a project, from defining the problem through to the implementation of co-created health solutions (Vargas, Whelan, Brimblecombe, & Allender, 2022).

To further clarify our definition of co-creation in this study, it is important to distinguish between co-creation, co-design and relevant design concepts — a task we have also covered in prior work (An, Sandlund, et al., 2023). Co-creation is an iterative process that involves various stakeholders and creative problem-solving; co-design, a specific instance of co-creation, involves collaboration between designers and non-designers (Masterson, Areskoug Josefsson, Robert, Nylander, & Kjellström, 2022; Messiha et al., 2023). Participatory design emerged in the 1960s in response to a growing demand for

greater consideration of community perspectives in major decision-making, emphasising the active involvement of end users in the design process and often being used interchangeably with co-design. Finally, design thinking places less emphasis on involving the end user and prioritises generating potential solutions as rapidly as possible (Luchs, 2015).

We have integrated strategies that cultivate appreciative intelligence among co-creators, enhance participant engagement, and ensure meaningful decision-making input. 'Appreciative intelligence' refers to the ability to reframe a given situation in order to recognise the positive possibilities that are embedded in it (Ghaye et al., 2008). Promoting appreciative intelligence among co-creators has been shown to aid in data collecting by allowing informants in health research to be viewed as equal stakeholders, and empowering them to engage positively with the subjects at hand (Ghaye et al., 2008; James, Blomberg, & Kihlgren, 2014; Tavemark, Kihlgren, Norell Pejner, & James, 2023). This shift in perspective may influence power dynamics in co-creation in public health contexts, making it especially valuable for engaging with populations with negative experiences related to the studied topics.

3 *Design methodology*

Integrating co-creation, Transition Design thinking, and appreciative intelligence resulted in a practical co-creation model for addressing public health issues. This model, primarily informed by the first two phases of Transition Design (Figure 2), has four stages − co−conception, co-envisioning, co-making, and co-mapping − for developing system-level health-intervention initiatives. The aims and actions for each stage are outlined in Table 1. To achieve the aims at each stage, the method choice should be tailored to the participants, cultural contexts, and other relevant situational factors.

4 *The complexity of hospitalisation issues in COPD care*

COPD is one of the most common chronic diseases, and the third most common cause of death globally (Lozano et al., 2012). The cost of healthcare for people with COPD is substantial, and the main driver for both increased healthcare costs and disease deterioration is acute exacerbation of COPD (AECOPD), which often requires hospitalisation (Ozkaya, Findik, & Atici, 2011). AECOPD is a clinical diagnosis that indicates that a person with COPD is experiencing worsened symptoms, leading to a change in medication and/or hospital admission (Agustí et al., 2023). Hospital routines have a proclivity towards early discharge for people with AECOPD, on the basis that this can free up hospital beds and lower the costs for the healthcare system (Chetty, MacKenzie, Douglas, & Currie, 2006). However, early discharge can increase the risk of readmission in the long run. Frequent rehospitalisation results in a high cost for the healthcare system, as well as challenges for people with COPD and their families (Iheanacho, Zhang, King, Rizzo, & Ismaila, 2020;

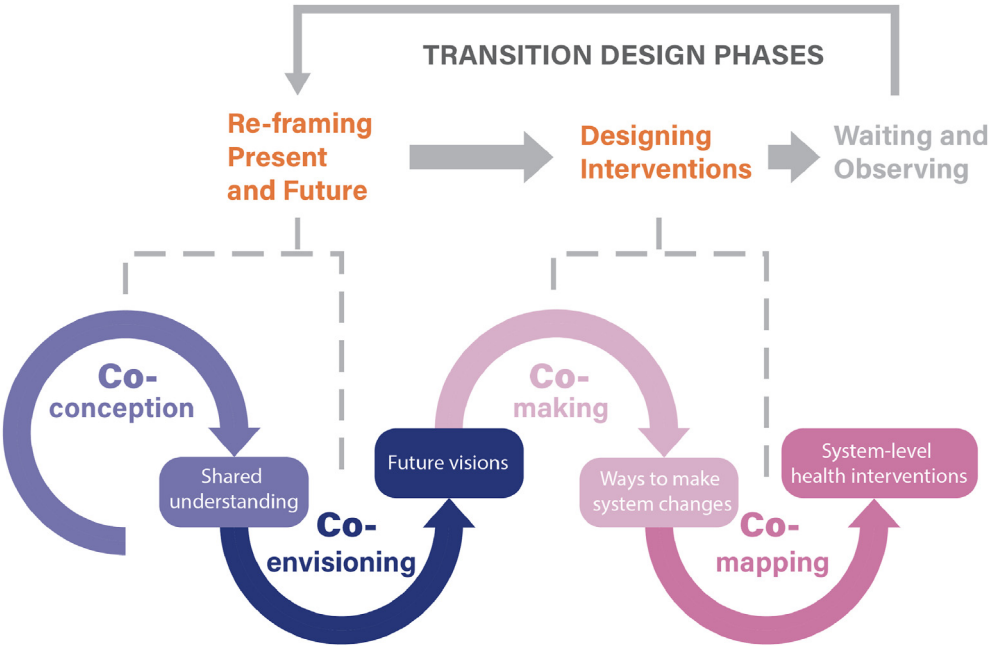


Figure 2 A Transition Design model for designing system-level health-intervention initiatives

Press, Konetzka, & White, 2018). Practices for addressing the challenges related to hospitalisation and early discharge of individuals with severe COPD often involve multidisciplinary teams; this involves collaboration between different healthcare practitioners, including nurses, physiotherapists, occupational therapists, and others with pulmonary rehabilitation expertise. Although interventions have been designed to address these challenges (Halpin, Miravitlles, Metzdorf, & Celli, 2017), the implementation of these frequently presents obstacles (Siu & Gafni-Lachter, 2024).

In our study, we treated the issue of hospitalisation and early discharge for individuals with COPD as a 'wicked problem'. Wicked problems are characterised by complex interdependencies between the nature of the issue and the potential solutions, making them particularly resistant to straightforward solutions (Rittel & Webber, 1973). The complex, dynamic, and often resource-poor nature of health systems makes them a fertile ground for the emergence of wicked problems (Ferguson, Hickman, Macbean, & Jackson, 2019). Addressing hospitalisation-related issues as wicked problems facilitates obtaining a deeper understanding of their complexity (Feo et al., 2023). By framing these as wicked problems in the context of COPD care, our study aimed to delve into this complexity, fully acknowledging its intricate nature as an essential starting point for deeper exploration and understanding.

Designing system solutions in hospital care

Table 1 The four stages of designing system-level health-intervention initiatives

Stage	*Aim(s)*	*Action(s)*
Co−conception	(1) To reframe how stakeholders see system difficulties so that they have an appreciation of their present status, and so that they have a positive perspective on how the future could unfold from the positive present. (2) To gain a shared understanding among stakeholders about the magnitude of the difficulty.	• Building an appreciative gaze (by e.g. asking appreciative questions). • Problem mapping (i.e. uncovering the causes and consequences of the problem). • Stakeholder mapping (i.e. uncovering the opposition and affinity relationships among the stakeholders).
Co-envisioning	To co-create a compelling future in which all of the needs of all stakeholders are fulfilled.	• Exploring individual preferred futures. • Generating preferred futures that co-creators yearn for.
Co-making	To identify the ways in which system changes can be made based on the results of the co−conception and co-envisioning stages.	Based on the current situation, identifying the ways (a term that we choose to describe the potential concepts that shape the initiatives) at various levels of care to achieve the co-created visions of the future.
Co-mapping	Based on the evolution of the system difficulties and the co-created visions of the future, to identify interdependent system-level health-intervention initiatives according to the links between them, with particular focus on development and implementation phases.	• Sorting the ways identified during the co-making stage based on whether they can support and enhance one another. • Generating tangible system-level health-intervention development initiatives based on the ways suggested. • Exploring how to accommodate the initiatives in the context of implementation. • Highlighting the links between initiatives for both the development phase (i.e. resources could be shared in the development phase) and implementation process. Regarding this latter aspect: 1. The prospective implementation facilitators for an intervention could meet those of another. 2. The implementation of one intervention could simultaneously be part of the solution to an implementation barrier for another intervention. 3. Certain aspects of one initiative could inform some of the implementation interventions for another intervention. Facilitators and barriers within an implementation process can be identified using the five domains of the Consolidated Framework for Implementation Research (CFIR): innovation domain, outer setting domain, inner setting domain, individual domain, and implementation process domain (Damschroder et al., 2009).

5 *Design experiment: Co-creating system solutions for COPD care*

This study is part of Health CASCADE, an EU-funded Marie Skłodowska-Curie Innovative Training Network that aims to make co-creation trustworthy and effective in the context of public health (Verloigne et al., 2023).

5.1 Study setting

The study was carried out at UMeHealth Lab at Umeå University, Sweden.

5.2 Participants

The study used purposive sampling to recruit participants from Norrlands universitetssjukhus ('University Hospital of Umeå'). The participants included people with severe COPD who had experienced hospitalisation in the preceding year due to acute COPD exacerbation, a family member of a COPD patient, healthcare practitioners (a physician, nurse, dietician, physiotherapist, and occupational therapist), a designer working with the eHealth programme design for the support and treatment platform 1177.se (the Swedish national healthcare service), hospital managers, and researchers. The researchers approached healthcare practitioners working in the multidisciplinary COPD management team at Norrlands universitetssjukhus regarding participation. People with COPD who met all of the eligibility criteria (Table 2) were approached by a COPD nurse at Norrlands universitetssjukhus and invited to participate in the study. The characteristics of the people with COPD and the family member are presented in Table 3, and those of the healthcare practitioners, the designer, and the hospital managers are presented in Table 4.

5.3 Co-creation practice

Table 5 contains descriptions of how the four stages of the proposed model were used in practice, with one 2-h workshop being performed for each stage. The questions posed during each workshop were framed appreciatively in order to help the participants view 'problems' from a positive and creative perspective (Ghaye et al., 2008). All of the workshop discussions were audio-recorded. After each session, the facilitators summarised the findings, together with the audio recordings, observational notes, and post-it notes made during the workshop; these then informed the next workshop. The intention was for the same group of participants to be involved in the first three workshops; however, due to the varying health status and availability of the participants, the number of participants at each session varied.

Table 2 Eligibility criteria for COPD patients

Inclusion criteria
• Diagnosed with COPD.
• Stable disease state, defined as no exacerbation in the month prior to recruitment.
• At least one hospitalisation due to COPD in the year prior to recruitment.
• Ability to read, write, and speak Swedish.
• Resident of the county of Västerbotten.
• Regular user of a tablet or smartphone.
Exclusion criteria
• Cognitive decline.

COPD: chronic obstructive pulmonary disease.

Table 3 Characteristics of the COPD patients and the family member

Characteristics	*COPD patients (n = 5)*	*Family member (n = 1)*
Gender (w/m)	3/2	1/0
Age (years)	66–82	N/A (not recorded)
FEV_1 predicted (%)	23–36	N/A
GOLD stage (n)	3E (3) 4E (2)	N/A
COPD assessment test[a] (min–max)	10–27	N/A
Communicative and critical health literacy scale (Swedish version)[b] (min–max/median)	5–1301/104	17
Electronic health literacy scale (Swedish version)[c] (min–max/median)	12–39/32	19

COPD: chronic obstructive pulmonary disease. w: women. m: men. FEV_1: Forced expiratory volume in 1 s. GOLD: Global Initiative for Chronic Obstructive Lung Disease.

[a] Scores ranging from 0 to 40, with higher values indicating worse symptoms.

[b] Scores ranging from 5 to 5000, with higher values indicating lower health literacy.

[c] Scores ranging from 8 to 40, with higher values indicating higher electronic health literacy. N/A: not applicable.

Table 4 Characteristics of the healthcare practitioners, designer, and hospital managers

Characteristics	*Healthcare practitioners and designer (n = 6)*	*Hospital managers (n = 3)*
Gender (w/m)	5/1	2/1
Age (years) (min–max)	25–61	46–62
Time in profession (years) (min–max)	0.8–33	22–34
Time working with COPD (years) (min–max)	0.8–23	5–15

w: women. m: men. COPD: chronic obstructive pulmonary disease.

5.3.1 Co–conception

Figure 3 is a photograph of self-portraits that were drawn by the participants when they arrived at the UMeHealth Lab. This activity was designed to enhance engagement and foster a sense of ownership among the co-creators. The participants were divided into three groups: allied healthcare practitioners and a family member; managers, a physician, a designer, and a nurse; and patients. This grouping was strategically planned in order to address the power dynamics between patients and healthcare practitioners, which can pose challenges to the co-creation process (Nimmon & Stenfors-Hayes, 2016). The same tasks were assigned to each group: first, to identify the causes of early discharge of AECOPD patients, and then to determine the consequences. Each group was instructed to write their findings down on post-it notes, as illustrated in Figure 4, which shows an example of the materials collected after the workshop. We adapted the World Café method to enable collaborative conversation and information-sharing (Brown & Isaacs, 2005). Bearing in mind the physical limitations of the patient group, the co-creators were not asked to move around, and instead the post-it notes were circulated in order to obtain consensus and gather feedback from all of the workshop

Table 5 The four stages of the proposed model, and how these were employed in practice

Stage	*Participants*	*Aim(s)*	*Method(s)*	*Outcomes*
Co−conception	(19 total) 5 COPD patients, 1 family member of a COPD patient, 5 healthcare practitioners (1 physician, 1 nurse, 1 occupational therapist, 1 physiotherapist, 1 dietician), 1 designer, 2 hospital managers, and 5 researchers[a]	To identify the causes and consequences of the problem of early discharge from hospital following AECOPD from different perspectives (political, infrastructure, social, economic, mental and physical wellbeing, and environmental).	World Café	Understanding of causes and consequences of the focused problem
Co-envisioning	(16 total)3 COPD patients, 1 family member of a COPD patient, 4 healthcare practitioners (1 nurse, 1 physician, 1 dietician and 1 physiotherapist), 1 designer, 2 hospital managers, and 5 researchers[a]	To co-create visions (in hospital/primary care, home environments, and community settings) of living with COPD in 2050, wherein the needs of various stakeholders are fulfilled.	Card game, cultural probes	Themes of the co-created future visions
Co-making	(12 total) 4 COPD patients, 2 healthcare practitioners (1 nurse and 1 dietician), 1 designer, and 5 researchers[a]	To collaboratively explore what could enable the transition to ideal futures from the present.	Provotype	Ways to achieve co-created future visions (see Appendix A)
Co-mapping	(5 total) 5 researchers[a]	To suggest system-level health-intervention initiatives based on the ways identified during the co-making stage, as well as to identify the sharable resources at the development stage and the synergies throughout the implementation processes of the initiatives.	Panel workshop	System-level health-intervention initiatives for addressing the problem (see Table 8); links between development processes and implementation processes for the initiatives (see Appendix B)

[a] Researchers: K.W., M.S., S.L., R.H., Q.A. COPD: chronic obstructive pulmonary disease.

participants. Inspired by thematic analysis, the data from the first workshop was summarised (Braun & Clarke, 2006). In addition, the participants were given a task to complete at home: answering the question "In an ideal world in 2050, how would you want things to operate regarding hospital care for COPD exacerbations?" This task was intended to be discussed at the next workshop.

Data from the first workshop identified the causes of early discharge following COPD hospitalisations: (1) Patients often felt neglected during their stay,

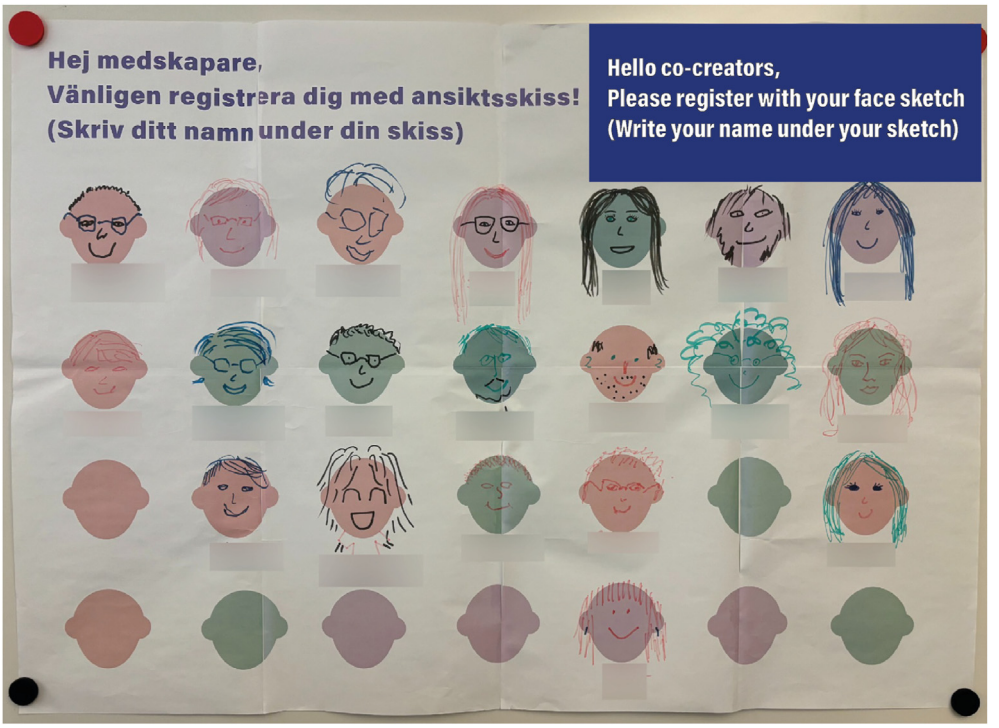


Figure 3 The participants drew themselves when they arrived at the first workshop (translation of the text provided in the blue box) (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

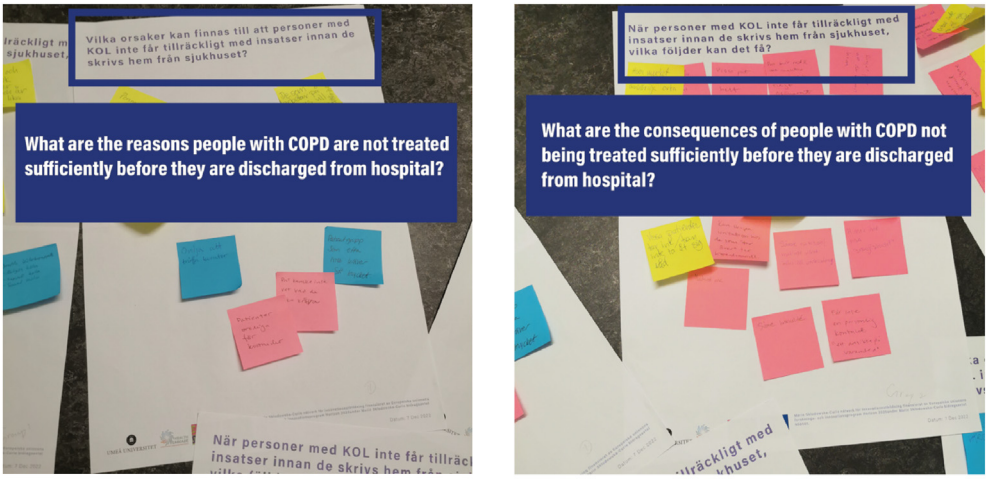


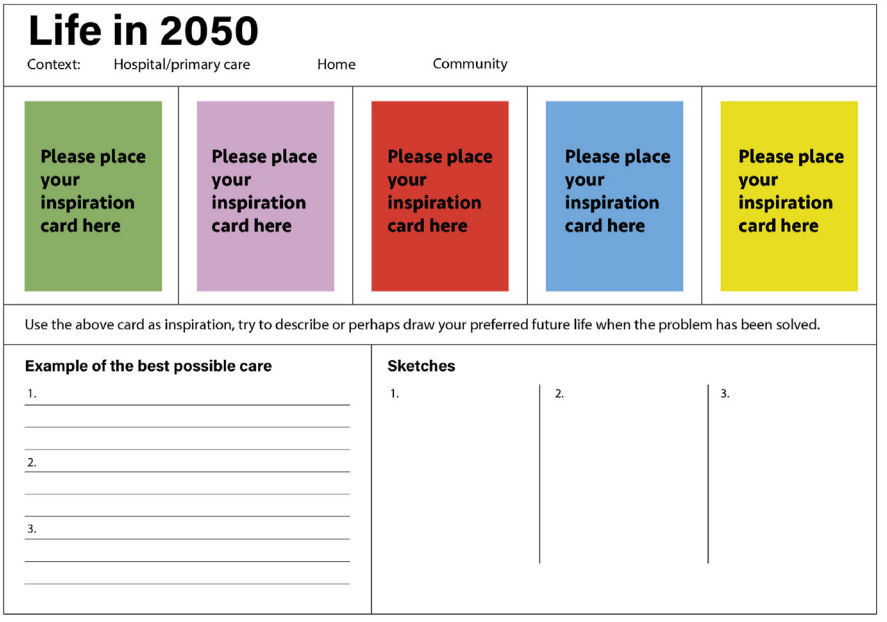
Figure 4 Examples of the templates used in the co−conception workshop (translation of the text provided in the blue boxes). (A) An example of the template used for the activity that explored the causes of problem. (B) An example of the template used for the activity that explored the consequences of the problem (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

possibly due to shortages of resources such as hospital beds and staff, discouraging prolonged stays. (2) A perceived lack of specific knowledge on COPD among healthcare practitioners, particularly those in initial contact with patients contributed to patients' anxiety regarding hospitalisation. Furthermore, patients observed poor interprofessional collaboration among the different practitioners in their care team, which resulted from their limited understanding of each other's roles. (3) Patients' limited knowledge about COPD hindered effective communication, allowing healthcare practitioners greater discretion in determining the timing of discharge. (4) Patients' desire for self-sufficiency could lead to resistance to extended hospital stays.

The consequences of early discharge were found to affect multiple stakeholders. For patients, it meant being discharged before meeting the full support team, leading to insecurity and guilt towards family members. It also limited trust-building with healthcare practitioners and left patients uninformed about post-discharge care options. Family members often faced stress and frustration due to the early discharge. For healthcare practitioners, frequent rehospitalisations resulting from early discharge led to job dissatisfaction and stress. Additionally, early discharge coupled with insufficient staffing can result in a system with a tendency to be overly dependent on individual personnel. For example, the assistance of nurses is critical to patients post-discharge, as patients are often still in poor condition and uncertain about their care following an early discharge. This reliance on nurses may increase their workload and stress.

5.3.2 *Co-envisioning*

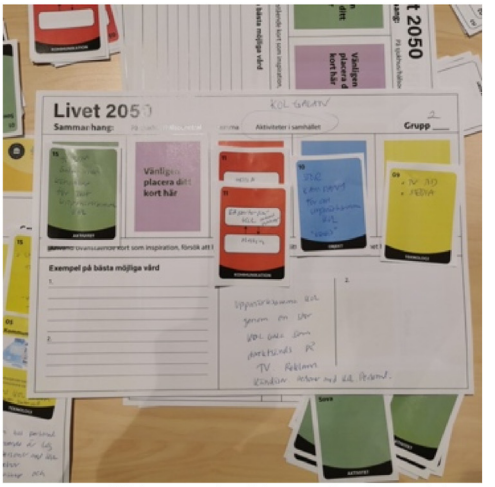
The goal of this workshop was for the co-creators to describe their ideal futures for the following settings: hospital/primary care, home environment, and community. The settings were chosen based on the data collected during the first workshop, in relation to the most impactful problems experienced during the discharge process. It is important to highlight that the topic of early discharge, which was introduced in the first workshop, was broadened during the participant discussions due to the complexity of the issue. As a result, subsequent workshops focused on a wider range of challenges relating to hospitalisation and early discharge. The workshop started with an education session on self-management strategies for people with COPD. Following a discussion about the home task from the first workshop, the participants were divided into five groups to co-envision their preferred future lives. To facilitate the brainstorming process, a collection of cards and card-game templates was provided for inspiration (Figure 5), adapted from the card game 'The Thing from the Future' (Candy & Watson, 2015). The cards were colour-coded to represent the different dimensions identified in sociotechnical systems (task, structure, technology, and people), with a set of blank cards included to increase flexibility (Leavitt & March, 1962). Each colour of card contained words



(A) *Template designed for the card game during the co-envisioning workshop (translated version).*



(B) The card game.



(C) An example of the completed material.

Figure 5 A template, setup, and example of the card game used during the co-envisioning workshop. ***(A)*** *Template designed for the card game during the co-envisioning workshop (translated version).* ***(B)*** *The card game. (C) An example of the completed material*

relating to a dimension, aiding participants in imagining playful and thought-provoking narratives regarding future sustainable sociotechnical systems for COPD care. Example narratives were provided to assist in scenario development. In addition, we devised a home task for the patients and family member that was informed by Cultural Probes method, which was intended to further investigate COPD-related problems at home (Merzali Celikoglu, Timur Ogut,

& krippendorff, 2017) (Figure 6). Table 6 outlines the aims of each tool included in the toolkit for Cultural Probes. The home task was intended to be discussed during the next workshop.

The co-envisioning workshop resulted in six visions for the future. The first of these was COPD patients avoiding unnecessary hospitalisations, with established home-care services taking precedence over hospitalisation. This shift would enable the pulmonary department to concentrate on patients who genuinely require hospital care and reduce the strain of treating patients in other departments. The second involved both patients and family members experiencing a sense of self-efficacy, self-esteem, and competence in self-management. In the third, care would be continuous, with no reliance on temporary healthcare practitioners who work brief stints in clinics; Electronic Health Records (EHRs) would be consistent, accessible, and easily understandable. In the fourth, there would be a sufficient number of healthcare

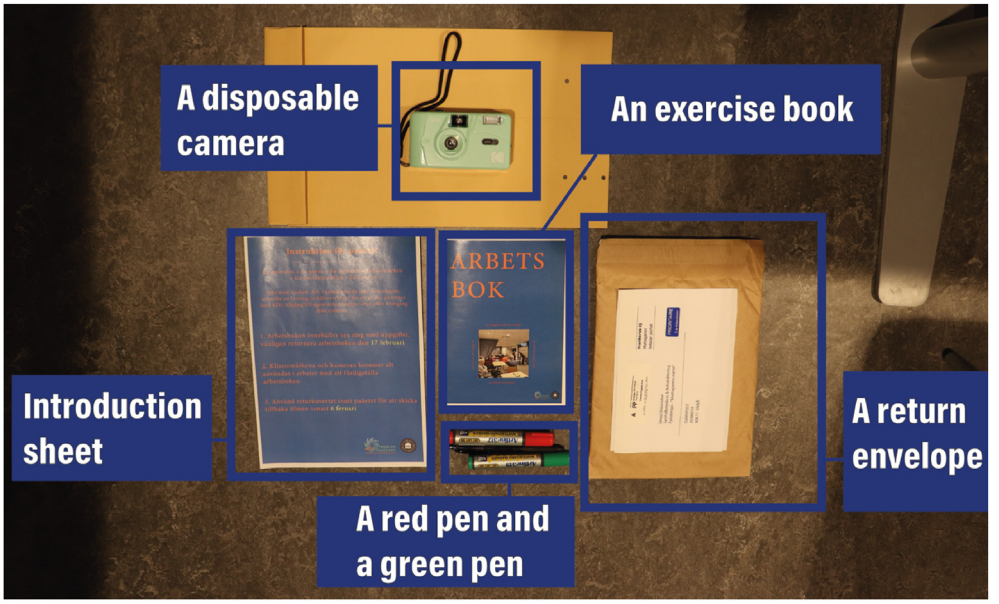


Figure 6 Photograph of the toolkit for Cultural Probes

Table 6 Toolkit for Cultural Probes

Object	*Aim*
An exercise book	To investigate the everyday routines of participants and the impact of COPD on their lives
A disposable camera	To determine what the participants believe constitutes safety
A red pen and a green pen	To explore perceptions of safety and danger
A return envelope	To facilitate the return of the exercise books and films
Introduction sheet	To provide instructions regarding how to use the items in the toolkit

Designing system solutions in hospital care

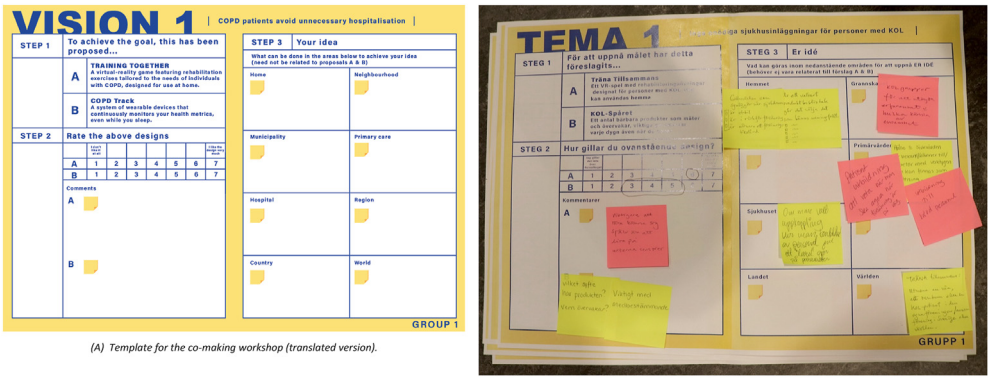
practitioners in the pulmonary department who feel satisfied with their work, and valued for their contributions. The fifth involved every COPD patient having a clearly designated contact person for advice, and ideally direct access to their assigned healthcare practitioners. In the sixth, general awareness of COPD would be heightened, contributing to both the prevention of the condition and improvements in the quality of care for those living with it.

5.3.3 Co-making

The participants were mixed and divided into three groups during the co-making workshop. Each was tasked with identifying ways of achieving two of the six visions generated during the co-envisioning workshop. An example of a template used for a vision is shown in Figure 7. We employed the 'provotype' method, which encourages conversation and stimulates designs by presenting provocative prototypes (Boer & Donovan, 2012). We suggested eHealth solutions based on each vision developed in the co-envisioning stage in order to inspire the co-creators to explore how to realise these envisioned futures. The provotypes, which were adapted from existing work and designed to include provocative elements, served as catalysts for discussion. For example, to address the idea of COPD patients avoiding unnecessary hospitalisations, a 'COPD Track' prototype was presented – "a system of wearable devices that continuously monitors your health metrics, even while you sleep". This was designed to provoke strong reactions and stimulate dialogue, encouraging participants to critically engage with the concept. The co-creators were asked to rate the provocative interventions, and offer feedback on whether each could help to realise the envisioned future. The ratings for the proposed interventions were low, as was expected given the intentionally provocative nature of each. They were also asked to suggest potential ways of achieving the six visions at multiple care levels (household, neighbourhood, municipality, primary care, hospital, region, country, and worldwide). To achieve these visions, 39 distinct ways at various levels of care were generated, based on the data collected from the co-making workshop; this was supplemented with data from the first two workshops (see Appendix A). In addition, the cultural probes task that the participants had completed at home was discussed in this workshop, offering researchers valuable insights into how COPD impacts the daily lives of patients.

5.3.4 Co-mapping

After the three workshops, the researchers gathered to suggest initiatives for system-level health interventions according to the five steps outlined in Table 7. Data from the co-mapping session were analysed alongside that from the preceding three stages of workshops and findings from existing research.



(A) Template for the co-making workshop (translated version).

(B) Photograph of an example of completed material.

Figure 7 Templates used in the co-making workshop. (A) Template for the co-making workshop (translated version). (B) Photograph of an example of completed material

Table 7 The steps followed during the co-mapping workshop for designing initiatives for system-level health interventions

Step	*Description*
1	Consider how the 39 distinct ways to achieve the six visions can support or enhance one another, and select the most promising ones for further development.
2	Generate tangible intervention-development initiatives based on these.
3	Identify the initiatives that could enhance one another to improve system performance.
4	Map the intervention-development initiatives (Figure 8). This frame details various aspects of intervention implementation, including the contexts in which the visions are to exist (the dimensions informed by the co-envisioning workshop) in relation to the contexts in which problems were raised (the dimensions informed by the co—conception workshop).
5	Produce a final matrix for the system solutions and discuss how those initiatives can enhance one another in the development and implementation phases. Further details are presented in Table 1.

The process resulted in the proposal of eight system-level health-intervention initiatives, primarily drawn from the 39 ways proposed for achieving the six visions (Table 8). These were mapped in a frame that outlines the implementation contexts (Figure 8). Detailed information regarding the links between the initiatives is provided in Appendix B.

5.4 Evaluation of co-creation process

Member checks and evaluation questionnaires were used to evaluate the co-creation process. Member checking involved the key results from each workshop being summarised and presented at the start of the next workshop, and confirmation that the results were correct was obtained from the co-creators. At the end of each workshop, the participants answered a questionnaire with seven-point Likert scales covering five dimensions of co-creative experience (hedonic experience, perceived knowledge, social atmosphere, attitude, and pragmatic experience) (Verleye, 2015). The primary goal of the questionnaire was to identify any areas for improvement, and to make timely modifications

Table 8 Descriptions of the proposed system-level health intervention initiatives (illustrated in Figure 8)

No.	*Initiative*	*Significance*
I	Development of digital and printed clinical checklists which can be used in various settings	Additional checklists used by healthcare practitioners working in primary care, ED, and pulmonary departments for COPD patients could help to enhance the quality of hospitalisation care by optimising the workflow of healthcare practitioners.
II	Development of digital educational resources for healthcare practitioners to learn about COPD, and how the condition is treated by other professions	Knowing what other professions treat COPD would allow healthcare practitioners to improve the quality of care by fostering better teamwork during the delivery process. Healthcare practitioners, especially emergency department staff, being more knowledgeable about COPD, could improve the admission experience for COPD patients and lead to more coordinated and comprehensive care.
III	On- and offline information campaigns	An information campaign both on social media and offline could increase public attention on the issues relating to the early discharge of COPD patients, which could result in more health services being offered.
IV	Development of digital and printed educational materials to improve knowledge of patients and family members regarding COPD	Patients prefer educational materials that are tailored to their needs and written in clear, accessible language. They would feel more secure if they could be able to assess their own symptoms.
V	Development of wearable devices to aid in the detection of potential exacerbations	These would facilitate better-informed decision-making about seeking professional help. Additionally, there is the potential for early detection of exacerbations, which could reduce the burden on emergency-care services.
VI	Development and implementation of routines and service	When symptoms appear, patients should be given a specific routine to carry out. The routine should be provided in an interactive and personalised way to improve the patient's experience.
VII	Integration of EHR systems	Enhancing the integration of EHR systems across different levels of health centre would provide continuity of medical records.
VIII	Development and implementation of services provided by a home-visit team	A home-visit team could facilitate a care plan tailored to the patient's individual circumstances, aligning with the vision of human-centred care. This approach could potentially reduce the need for hospitalisation and rationally achieve early discharge.

COPD: chronic obstructive pulmonary disease. EHR: electronic health record. ED: emergency department.

to the subsequent workshops. The results of the evaluation questionnaires are shown in Table 9. The co-creators rated the workshops highly across all dimensions; perceived knowledge was rated lower than other dimensions after the first workshop, but increased subsequently as a result of an education session being included at the start of the other workshops.



Figure 8 The implementation contexts for the system-level health interventions generated during the co-mapping session. (Each Roman numeral represents a system-level-health intervention initiative presented in Table 8.)

Focus-group discussions were then conducted to evaluate the participants' experience of taking part in the co-creation project. The co-creators were divided into three groups, and three external facilitators who had not previously interacted with any of the co-creators were recruited to lead the focus groups. The three facilitators were briefed on earlier workshops, and informed of the questions we sought answers to. The question guide contained six questions (see Appendix C). Overall, the co-creators were satisfied with the co-creation practice. The healthcare practitioners mentioned that it was great to see the patients in a stable condition outside the hospital. The co-creators generally felt excited and had learnt a lot by hearing different perspectives. Some suggested that it would be beneficial to include the viewpoints of politicians during the co-creation process. The co-creators believed that the tasks were valuable, but also expressed a need for additional time for thorough discussion. Additionally, some co-creators mentioned feeling limited in terms of their creativity during some of the tasks, especially the co-envisioning workshop. They felt that it was great to envision the future, but that there was limited creativity for imagining a preferred future life. They especially struggled to anticipate future technological advancements due to their limited experience of digital tools, which made it particularly challenging to include technical elements in their future visions. Finally, some noted that an

Table 9 The results of the evaluation questionnaires

	1st Workshop Median (min–max) (n = 13)	*2nd Workshop Median (min–max) (n = 9)*	*3rd Workshop Median (min–max) (n = 5)*
Statement 1 (hedonic experience): I enjoyed participating in the workshop	6 (3–7)	7 (6–7)	7 (6–7)
Statement 2 (perceived knowledge): I gained new knowledge/skills from the workshop	5 (1–7)	5 (4–7)	6 (5–7)
Statement 3 (social atmosphere): I feel that my opinion was valued as much as those of others	7 (4–7)	7 (5–7)	7 (6–7)
Statement 4 (attitude): I believe it is beneficial to include end users and other stakeholders in the design process	7 (5–7)	7 (6–7)	7 (6–7)
Statement 5 (pragmatic experience): I was able to contribute to the discussions during the session	7 (4–7)	6 (4–7)	7 (7–7)

Likert scale, 1: don't agree at all, 7: totally agree.

extroverted and articulate co-creator was essential to starting the conversations and stimulating group creativity.

5.5 Facilitator experience

The facilitators discussed their experiences of the model. Transcripts of these discussions were analysed, and the results of this analysis were categorised into positive and negative groups, which were then validated with the facilitators. They found the stages of the proposed model to be a clear roadmap that provided good direction for the process. They emphasised the need to balance strictness and flexibility when applying the model, as being too rigid could impede the co-creation process. Additionally, they described decentralised power relations among the co-creators, especially between patients and healthcare practitioners, and felt that this was effective for uncovering implicit information, such as perspectives on one another's behaviour during hospitalisation.

Recommendations for further improvements to the model were also made. The first of these was that additional workshops would generally benefit all stages by revealing more knowledge about each stage. Second, external researchers could have been beneficial for the co-mapping stage, as external perspectives can offer valuable insights regarding project ideas and add to the interdisciplinarity of the stakeholder perspectives. Third, the facilitators advised against asking too many questions during a single task and requesting that participants come up with detailed and specific solutions. Fourth, the tasks to be completed at home produced numerous and valuable outputs that were underutilised. For example, one of the home tasks was based on

the idea of Cultural Probes, and was intended to help the research group gain a better understanding of the daily lives of patients, resulting in greater empathy. The task revealed several dimensions relating to challenges in daily life, but some critical aspects of their demands were not addressed, either through co-created outcomes or other means. The facilitators felt it would have been better to clarify the goals of the home tasks that were given following each stage.

6 *Discussion*

The research presented in this paper involved applying a model primarily based on Transition Design in order to address a public health problem. By building on the Transition Design approach, the model integrated the principles of co-creation and appreciative intelligence in public health, successfully adapting these concepts to systemic problem-solving in the public health context. Taken together, the stages of the model provided a clear roadmap for exploring system solutions according to the facilitators. The co-creators felt involved and appreciated, and that their participation was valuable. The application of the model resulted in system-level health-intervention initiatives that can improve COPD hospitalisation care. The outcomes of system-level health interventions could be particularly advantageous to large-scale project initiatives that address complex public health issues. Multidisciplinary aspects and internal linkages in the development and implementation process characterise system-level health interventions generally, creating a system (ecology) wherein initiatives can reinforce one another. This discussion, organised into several sections, addresses three key questions: 1) How does this model distinguish itself from other participatory and systems-thinking methodologies? (Section 5.1); 2) What are the unique challenges and benefits of applying Transition Design in public health, as compared to other fields? (Section 5.2); and 3) What implications does this study have for design researchers working on system solutions in healthcare contexts (Section 5.3)?

6.1 *Differentiating the proposed model from other participatory and systems-thinking methodologies*

Participatory design workshops are widely recognised for their ability to address public health problems. Various tools and principles have been developed to aid designers involved in participatory design processes (Heiss & Kokshagina, 2021; Hendriks, Huybrechts, Slegers, & Wilkinson, 2018). However, a persistent challenge remains in empowering both the providers and recipients of care to actively contribute to and advocate for health solutions (Groeneveld, Dekkers, Boon, & D'Olivo, 2018). Our model theoretically addresses this challenge by incorporating co-creation principles in the context of public health, facilitated through an appreciative approach that decentralises the designer's role and redistributes decision-making power among all co-creators. To address the risk of tokenism in the co-design process, it is crucial

to rethink the role of designers (Lee, 2008). Our study builds on this insight by redefining the designer's role in order to focus on co-creation planning, iterative data analysis, and rapid prototyping. In practice, this involved adapting methods at each workshop stage, synthesising key messages from co-creators for validation in subsequent sessions, and rapidly prototyping co-creators' needs, primarily in written formats. Our approach aligns with Lee's perspective, supporting the evolution of designers into roles such as design developers, facilitators, and generators (Lee, 2008).

Some systems-thinking methodologies are already utilised in public health research, both for designing and evaluating interventions by elucidating their interactions with broader systems (McGill et al., 2021; Peters, 2014; Rusoja et al., 2018). The proposed model may enhance these methodologies in several ways:

a) It scaffolds the synergy-development process between system solutions by facilitating identification of connections across development and implementation phases. This structured approach enables the creation of system-informed implementation plans for interconnected interventions, grounded in a shared understanding of the problem among co-creators. By doing so, it seeks to tackle the challenges associated with implementing systems interventions (Whelan et al., 2023).
b) The model emphasises the importance of exploring stakeholders' lived experiences, addressing a common criticism of systems thinking: that it often overlooks the daily experiences of individuals due to limited stakeholder involvement or participation (Buchanan, 2020; Movsisyan, Rehfuess, & Norris, 2020). Developing a system perspective among participants allows them to participate meaningfully, which is important in designing system interventions (Buchanan, 2020). Co-creation facilitated the adoption of a systems perspective by participants, and has the potential to foster internal interaction among co-creators, resulting in the development of shared understanding and mutual sympathy.
c) The model critically examines existing unsustainable socioeconomic and political paradigms. It emphasises the importance of balancing future aspirations with present realities, recognising that an exclusive focus on current contexts may hinder the development and scaling of innovative solutions (Irwin, 2020; Proctor et al., 2021). While some may view a future-focused approach as impractical, or as potentially detracting from actionable insights, the model integrates insights from envisioned futures with current conditions to foster more deliberate and sustainable transformation (Fazey et al., 2020). This approach aligns with a growing trend among systems researchers and organisations in exploring future-oriented problem-solving to drive deliberate and sustainable change (Keseru, Coosemans, & Macharis, 2021; Neuhoff, Simeone, & Laursen, 2023; Sools, Groot, Coppers, & Triliva, 2022).

6.2 *Positioning transition design in public health research*

Irwin highlights the importance of designers expanding their capabilities by integrating systems thinking across diverse domains (Irwin, 2015). To our knowledge, the research presented in this paper is the first systematic attempt to apply Transition Design thinking to the issue of the hospitalisation among a patient group with a severe disease. The use of Transition Design in health research presents both notable challenges and unique opportunities.

One major challenge is addressing health equity, as Transition Design often involves designing for marginalised populations who face barriers relating to socioeconomic status, race, or geographical location. It is essential to gather data on the social determinants of health, and to identify the systemic inequities that affect these communities (Taylor et al., 2016). To explore these aspects, our study involved researchers with expertise in interacting with COPD populations in Northern Sweden. We also found that health equity for advanced-stage COPD patients could be influenced by disparities in digital literacy, with older patients and those with certain comorbidities often having limited access to digital technology. The patients in our study were somewhat resistant to some advanced health technologies and emphasised the importance of human interaction during times of suffering. Further research into health problems that uses Transition Design may benefit by adopting a cautious stance on technology, which may mitigate mistrust among participants and reinforce the importance of empathetic, human-centred care. Another practical challenge in applying Transition Design in healthcare is managing the power dynamics between patients and healthcare practitioners (An, Sandlund, et al., 2023). To ensure fair participation, we employed warm−up activities during the first meeting to encourage co-creators to step outside their identities through role-playing games, framed questions in an appreciative manner and included more patients than healthcare practitioners in each discussion. Furthermore, while hospital managers may not interact directly with patients, their deep understanding of the care-delivery process greatly enhances the co-creation process, ensuring that interventions are well-integrated and effectively address broader system dynamics.

Despite some resistance to new technologies on the part of the participants, our findings suggest that Transition Design may support prudent digitalisation in health interventions. To encourage the sustained use of eHealth interventions, their design should be considered alongside other interventions within the healthcare system, in order to prioritise system demands over technological improvements (An, Kelley, Hanners, & Yen, 2023; Ferati et al., 2022). While technological progress is crucial for modern eHealth interventions, this must align with the capabilities of the existing healthcare system, and be driven by market demand (Cerrato & Halamka, 2019; Greenhalgh et al., 2015; Safi, Thiessen, & Schmailzl, 2018; Teyhen et al., 2014). The

proposed system-level health-intervention initiatives include project ideas for eHealth intervention development, and offer a way of causing eHealth interventions to interact with other interventions through identified linkages. Such synergies may create more practical intervention designs and implementation strategies, with higher likelihoods of sustainment in the system. Additionally, the adaptable structure of Transition Design fosters a multidisciplinary approach to problem-solving, as demonstrated by the diverse dynamics of the initiatives it supports. This adaptability allows the limitations of current public health paradigms, which frequently depend on uniform frameworks, to be addressed, and creates opportunities for cross-disciplinary funding (Trochim, Cabrera, Milstein, Gallagher, & Leischow, 2006).

6.3 *Implications for system solutions in the context of public health*

Our investigation resulted in four key implications relating to the application of the model to healthcare settings, or employing systems thinking to collaboratively address problems with key stakeholders, being identified.

1) **Enhancing future visions:** Our study highlighted challenges in conceptualising future visions within healthcare. We found that some anticipated futures were too abstract or visionary, or lacked concrete detail. Although 'bold visions' are encouraged during this forecasting process, as is noted in Irwin's (2020) work on Transition Design, it is also beneficial to make the future more tangible and practical, especially considering that the issues we were addressing were relatively localised and specific. Although methods used in the co-envisioning stage, such as the card game, may have imposed certain constraints by suggesting existing items on the cards and thus making visions more practical, incorporating additional strategies could be advantageous. If resources permit, integrating insights from Ethnographic Experiential Futures (EXF) may further enhance this stage (Candy & Kornet, 2019). EXF combines ethnographic and experiential foresight, facilitating engagement with diverse participants and iterative refinement of future scenarios based on feedback, leading to more actionable outcomes (Candy & Kornet, 2019).
2) **Involving co-creators in method adaptation:** Although the co-creators generally expressed satisfaction with the workshops, they experienced some limitations in terms of creativity, which could have affected their sense of ownership (Olszewski, 2022). These limitations may have arisen from insufficient time allocated for the tasks or from the lack of clearly defined exploration opportunities, possibly resulting from the adaptation of methods across different stages. Future method selection could benefit from collaborative decision-making with co-creators to enhance their sense of ownership.

3) **Addressing retention challenges:** The co-creation process is time-intensive, making it challenging to maintain co-creator engagement throughout workshops. To improve retention, we recommend increased training and upskilling activities to improve motivation among participants (Alkumaim, Alhazmi, Ramayah, Shabbir, & Gazem, 2021; Lundell, Toots, Sönnerfors, Halvarsson, & Wadell, 2022). We introduced an educational session on self-management that was delivered by one of the facilitators based on knowledge gaps identified through feedback questionnaires, and this significantly enhanced the learning experience of the participants in subsequent workshops. Building on this experience, we suggest incorporating upskilling activities led by co-creators in workshops, in order to further enhance interactions and address power imbalances. For example, healthcare practitioners could lead educational sessions where appropriate. Moreover, the retention rates among healthcare practitioners were lower than those among patients in our study, indicating a potential need for improvement. The healthcare practitioners in our study reported that engaging with patients in non-hospital settings provided them with valuable insights, which enhanced their empathy and understanding. So we believe methods that facilitate deeper interaction between co-creators could increase the motivation and commitment to ongoing participation of healthcare practitioners in the project. Therefore, future adaptations of the model may benefit from including activities that enhance interaction, enrich knowledge, and offer new insights for all participants.
4) **Adapting initiatives for broader use:** One of the sustainable uses of system-level health-intervention initiatives is adapting them to new contexts, particularly environments with limited resources for undertaking co-creation activities (Moore et al., 2021). However, our study, along with existing literature, shows that co-created outcomes tend to be localised, which can limit their generalisability and adaptability to other settings (Leask et al., 2019). To increase the likelihood of outcomes being generalisable and adaptable, it is important to clearly document the design process and adhere to recognised guidelines regarding transparency. Employing frameworks such as PRODUCES framework (PRoblem, Objective, Design, Users, Co-creators, Evaluation, Scalability) for co-creation practice reporting can enhance documentation and facilitate the transferability of initiatives to new contexts (Leask et al., 2019).

6.4 *Limitations*

First, data scalability may have been an issue due to the small sample size: only five individuals with COPD in the co-creation sessions. This problem could be ameliorated by involving healthcare practitioners with expertise in managing diverse patient scenarios. However, since the optimal number of participants for a co-creation process is typically 10–12 (Leask et al., 2019), our sample

size of patient group may actually have been ideal for the intended outcomes. Second, additional workshops could yield more comprehensive results, but their cost may warrant exploration of digital alternatives, which have been shown to be effective in similar contexts (Lundell et al., 2022). Third, the co-creators did not provide input on the final proposed system-level health-intervention initiatives, as the co-mapping session necessitated a thorough and extensive analysis. Fourth, during the co-mapping session, while the design decisions regarding system-level health-intervention initiatives were primarily made by a team of researchers from COPD care and design disciplines, there was a missed opportunity to incorporate external perspectives from researchers in fields such as informatics. Introducing these perspectives could have enhanced the interdisciplinary collaboration and novelty of the ideas for the initiatives. Fifth, this study was limited by project timelines and reporting constraints, preventing a comprehensive exploration of Transition Design. Transition Design is a systems-thinking methodology that focuses on finding long-term solutions in order to bring about system changes. This involves observing how a system reacts to interventions on the basis that, as is noted in other research that has made use of Transition Design, establishing routine evaluations of outcomes is important (Iwabuchi & Mizuno, 2020). By the time of submission of this paper, some of the proposed system interventions will have been developed. The intention is to test these in a feasibility study and investigate how the system responds to the interventions.

7 *Conclusion*

This study investigated the application of a model primarily grounded in Transition Design to address the complexities of healthcare challenges in the COPD domain, offering practical evidence for its broader use in other complex healthcare contexts. The findings suggest that the model is useful as it gave input on, and suggestions for, how to enhance implementation effectiveness and sustainment of co-created system-level health interventions. The integration of co-creation and appreciative intelligence within the model was instrumental in building a shared understanding of the challenges and facilitating systems thinking among the workshop participants.

Importantly, this study highlights the benefits of decentralising the designer's role, transforming the process into a collaborative and interpretive endeavour shared between co-creators and designers. Empowering participants with greater decision-making authority enhanced the responsiveness of the outcomes to multiple needs but also introduced new challenges, especially during the forecasting and intervention design phases. These challenges included the added creative responsibility that comes with greater decision-making power and the difficulty of ensuring the feasibility of envisioned futures and proposed interventions. Addressing these obstacles may benefit from more tailored, co-

developed methods that engage stakeholders and enhance the feasibility and impact of resulting visions and initiatives.

Although this study was an initial exploration of problem-solving and further testing is required, it provides novel insights for design researchers who seek to engage with stakeholders in the co-creation of system health solutions. The structured design of system-level health-intervention initiatives will aid in the evaluation of system improvements in the following stage of this project.

Future research could strengthen this model by developing more detailed criteria for the methods used at each stage and establishing more rigorous evaluation processes for the outputs. Complex healthcare challenges could benefit from integrating the model earlier in the problem-solving process, allowing for a more comprehensive understanding of systemic complexities. This approach seeks to increase the potential for sustainable, long-term impact by revealing connections between key intervention points.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Qingfan An: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Marlene Sandlund:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sara Lundell:** Writing – review & editing, Resources, Investigation, Formal analysis, Data curation. **Christoffel Kuenen:** Writing – review & editing. **Sebastien Chastin:** Writing – review & editing, Formal analysis. **Ragnberth Helleday:** Writing – review & editing, Investigation, Formal analysis. **Katrina Messiha:** Writing – review & editing, Formal analysis. **Quentin Loisel:** Writing – review & editing, Formal analysis. **Karin Wadell:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical statement

The study was performed in accordance with the Declaration of Helsinki and was approved by the Swedish Ethical Review Authority (Dnr. 2022-03695-01). Signed informed consent was collected from all participants before the start of the study.

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Appendix A−C. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.destud.2025.101297.

Data availability

Data will be made available on request.

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